# FLOOD INTELLIGENCE 3D

## Physics-Informed Satellite Earth Observation & Hydrodynamic Flood Inundation Engine

### Project & Submission Details

- Theme: Disaster Management / AI for Social Good (HSC Hackathon)
- Concept: Dual-Polarization SAR + Physics-Informed Neural Network (PINN)
- Team Name: [Your Registered Team Name]
- Team ID: [Your Registered Team ID]
- State / UT: Tamil Nadu (or your registered state)
- Target Stakeholders: NDRF, SDRF, GDACS, Copernicus EMS, Disaster Relief Forces
- Primary Goal: Reducing disaster response lag from 48 hours to under 60 seconds

### Live Prototype & Open-Source Artifacts

- 24/7 Production Live Web App: https://flood-intelligence-3d-cy4j.vercel.app
- Official GitHub Repository: https://github.com/gokulapriyan1119-dev/flood-intelligence-3d
- Technical Stack: Next.js 14 App Router · React Three Fiber · Three.js · TailwindCSS
- Inference Pipeline: U-Net++ (ResNet-34) · Manning Navier-Stokes Hydrodynamic PINN
- Production Security: Enterprise Google SMTP 2FA Email OTP Verification System
- Scientific Outputs: Automated 10-Page PDF Reports & Spec-Compliant TIFF 6.0 Rasters
- License: Official Open-Source MIT License

Team: [Your Team]

# Problem – Proposed Solution

**Hackfor**
**Social Cause**

## 1. Core Problem

- Optical Satellites Blinded: Heavy monsoons, cyclones, and nighttime clouds completely obscure standard optical satellites (Sentinel-2, Landsat).
- Static 2D Boundaries Only: Conventional flood maps only show water presence/absence without continuous water depth (h).
- Fatal Rescue Bottlenecks: Relief boats and vehicular convoys run aground in unseen shallow ridges or drown in unexpected deep pits.
- 48-Hour Response Lag: Manual GIS processing takes days, missing the vital golden hours of emergency response.

## 2. Proposed Solution

- All-Weather C-Band Radar: Ingests Sentinel-1 active radar backscatter (sigma0 in dB) to pierce storms and cloud cover 24/7.
- Dual-Stage AI Engine: U-Net++ (ResNet-34) segments flood extent; PINN solves 2D Shallow Water PDEs for continuous depth.
- Manning Topographic Coupling: Combines 30m Copernicus DEM elevations with physical channel surface roughness equations.
- Interactive 3D WebGL Command Center: Interactive rotating 3D Earth & hydrodynamic bathymetry terrain directly in browser.

## 3. Innovation & Uniqueness

- Physics-Constrained AI: Enforces Navier-Stokes mass and momentum conservation in loss functions.
- Dynamic Cross-Section Slice: Cuts interactive 2D bathymetric elevation profiles across terrain in real time.
- 1-Click 10-Page PDF Reports: Instant generation of military-grade flood situation dossiers via jsPDF.
- Zero Desktop Software Cost: 100% accessible on standard smartphones without costly ESRI/ArcGIS licenses.

Team: [Your Team]

# Technical Approach & Architecture



## Technologies Used

- Frontend & 3D: Next.js 14 App Router, TypeScript, React Three Fiber (R3F), Three.js, TailwindCSS.
- Data Viz: Recharts (hydrographs), Framer Motion, Lucide Icons, Canvas-Confetti.
- Backend & ML: FastAPI, PyTorch, U-Net++ (ResNet-34 Encoder), Hydrodynamic PINN.
- Remote Sensing: Sentinel-1 C-SAR IW GRDH (VV/VH), Copernicus DEM GLO-30, ECMWF ERA5.
- Security: Nodemailer + Google SMTP (2FA Real-Time Email OTP Verification).
- Deployment: Vercel Global Edge (24/7 Cloud), Docker, Cloudflare Tunnel.

## Methodology & Pipeline Process

- 1. Data Ingestion: Drag-and-drop dual-pol GeoTIFFs (VV/VH radar backscatter in dB).
- 2. Radiometric Preprocessing: Sigma0 calibration + Refined Lee 7x7 speckle filter.
- 3. Deep Segmentation: U-Net++ extracts water-land inundation mask (Test IoU: 0.621).
- 4. PINN Hydrodynamic Depth Solver: Enforces dh/dt + div(uh) = R(t) alongside Manning friction laws (Residual <= 4.18e-5).
- 5. 3D WebGL Rendering: Displaces terrain mesh with Copernicus DEM elevations & animated water wave shaders.
- 6. Multi-Format Reporting: Generates 10-page institutional PDF situation reports & spec-compliant TIFF 6.0 RGB rasters.

## AI Tools & Links

- AI Usage Extent:
- - PyTorch U-Net++ for radar backscatter segmentation.
- - PINN solving 2D Shallow Water differential equations.
- - Automated situational report text generation.
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- Repository Link:
- github.com/gokulapriyan1119-dev/flood-intelligence-3d
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- Live Production URL:
- flood-intelligence-3d-cy4j.vercel.app

# Impact Potential & Societal Value



## Impact on Target Audience

- Sub-Minute Situational Awareness: Slashing disaster analysis latency from 48 hours to under 60 seconds.
- Safe Rescue Navigability: First responders receive pinpoint depth maps (safe roads vs flooded culverts), preventing vehicle groundings.
- Critical Facility Triage: Identifies vulnerable hospitals, power substations, and bridges before submersion.
- Democratized Emergency Access: Zero installation barrier for field volunteers and relief coordinators.

## Social & Economic Benefits

- Drastic Mortality Reduction: Continuous depth warnings protect high-density flood plains (e.g., Assam Brahmaputra, Bihar, Godavari).
- Multi-Million Asset Preservation: Enables authorities to build sandbag barriers around transformers and substations in advance.
- Targeted Relief Airdrops: Directs rations and medical supplies to genuinely isolated depth zones rather than generalized flooded areas.
- Zero Commercial Licensing Cost: 100% open-source alternative to expensive commercial GIS software suites.

## Primary Stakeholders

- Disaster Management Forces: National & State Disaster Response Forces (NDRF, SDRF, SDMA).
- International Humanitarian Agencies: UN OCHA, Copernicus EMS, GDACS, Red Cross.
- First Responders: Coast Guard, Navy rescue squads, state fire and rescue teams.
- Urban Local Bodies: Smart city drainage boards and municipal flood control cells.

# Feasibility and Risk Analysis



## 1. Feasibility Assessment

- Technical Feasibility: Fully functional production prototype deployed on Vercel Edge with zero build or runtime errors.
- Data Feasibility: Relies entirely on free, open-access Copernicus Sentinel-1 radar and ESA DEM datasets.
- Economic Scalability: Serverless architecture handles concurrent users with zero infrastructure maintenance expenses.
- Hardware Agnostic: Client-side WebGL renders 3D bathymetry smoothly on any standard laptop or mobile browser.

## 2. Potential Challenges & Risks

- Urban Radar Glare: Double-bounce reflections off buildings can mask water in high-density metropolitan zones.
- Satellite Revisit Cycles: Sentinel-1 orbital paths provide 6-to-12 day revisit intervals.
- Crisis Network Outages: Severe cyclones and flooding can disrupt local cellular towers and field internet.
- High-Resolution Processing: Large satellite rasters can strain computational memory.

## 3. Mitigation Strategies

- Refined Lee 7x7 Filtering: Dual-polarization ratio (VH/VV) calibration eliminates urban corner noise.
- PINN Hydrodynamic Simulation: Forward-propagates flood waves using ECMWF ERA5 rain data between satellite passes.
- Offline-First Export: Standalone 10-page PDF reports and TIFF 6.0 rasters viewable offline on any field tablet.
- Optimized Window Tiling: Processes large satellite rasters in memory-optimized 512x512 patches.

Team: [Your Team]

# Team Composition & Journey



## Team Composition & Roles

- Gokulapriyan (Lead Architect):
- - Fullstack Next.js 14 architecture, Three.js/R3F 3D earth and bathymetry shaders, Vercel cloud deployment, authentication.
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- Geospatial & Remote Sensing Specialist:
- - Sentinel-1 SAR polarimetric calibration, Copernicus DEM GLO-30 elevation integration, GeoTIFF binary pipeline.
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- Hydrodynamics & ML Engineer:
- - U-Net++ ResNet-34 segmentation, Manning 2D Shallow Water PINN loss formulation.

## Technical Challenges Overcome

- Sensor-Free Depth Modeling: Calculated continuous depth without ground sensors by enforcing Navier-Stokes physics in neural loss.
- 60 FPS 3D Rendering: Maintained smooth frame rates in browsers via custom vertex displacement shaders.
- Hybrid 2FA Security: Built real-time Google SMTP OTP for new users with friction-free logins for returning sessions.
- Spec-Compliant TIFF 6.0: Developed binary image writer readable in standard Windows Photo Viewer and GIS software.

## Journey & Key Learnings

- Theory to Operations: Converted complex academic fluid dynamic theory into an operational emergency humanitarian tool.
- Rapid Prototyping: Iterated, tested, debugged, and cloud-deployed a production-grade system within hackathon timelines.
- Social Impact Commitment: Dedicated to empowering disaster relief agencies with life-saving open-source technology.

# References & Open-Source Artifacts



## Academic, Scientific & Project References


- 1. Copernicus Emergency Management Service (EMS): Sentinel-1 SAR Flood Inundation Delineation Protocols and Water Backscatter Thresholding Guidelines.
- 2. Zhou, Z., Siddiquee, M. M. R., Tajbakhsh, N., & Liang, J. (2018): "UNet++: A Nested U-Net Architecture for Medical and Satellite Image Segmentation." IEEE Transactions on Medical Imaging.
- 3. Raissi, M., Perdikaris, P., & Karniadakis, G. E. (2019): "Physics-Informed Neural Networks: A Deep Learning Framework for Solving Forward and Inverse Problems Involving Nonlinear Partial Differential Equations." Journal of Computational Physics, 378, 686-707.
- 4. European Space Agency (ESA): Copernicus DEM GLO-30 Global Digital Elevation Model (30m Resolution Topography).
- 5. European Centre for Medium-Range Weather Forecasts (ECMWF): ERA5 Hourly Atmospheric Precipitation Reanalysis Datasets.
- 6. Official Project Code Repository: https://github.com/gokulapriyan1119-dev/flood-intelligence-3d (Open-Source MIT License).
- 7. 24/7 Global Production Web Application: https://flood-intelligence-3d-cy4j.vercel.app (Live Deployment).